Faithfulness and the maps with at most two flag orbits

Human Compression project

Working proof edition; independent review pending

Abstract

We recover the principal faithfulness, symmetry and mixing results of Cunningham, Mochán, Montero, Potočnik and Toledo by applying the common theory of equivalence relations and transporters. In rank three, unfaithfulness reduces to one of two endpoint identifications. The remaining classification is a calculation in a dihedral group, including infinite polygons. The final construction in the source needs a change of one generator in its displayed formula. Its converse also requires attention to a degenerate choice of base generators. The corrected construction and the finite example are stated separately from compression of valid results.

1 Convention and source scope

Write $I = \{0, \dots, n-1\}$ and use the definitions in *Equivalence relations, transporters and flags*, cited below as the common paper. Color words act on the right. Put $F_i(x) = xW_{I\setminus\{i\}}$ and $C(x) = \bigcap_i F_i(x)$. Thus $x \sim y$ means that x and y determine the same maximal chain of faces. The symbols for involutions below refer to actions on flags; when an automorphism is prescribed by its value at a base flag, it is distinguished from the color connection having that value.

The source, abbreviated [F], is *Faithfulness in maniplexes*, Journal of Algebraic Combinatorics 64 (2026), article 23, doi:10.1007/s10801-026-01559-y. Our exact edition has 29 physical pages. The principal contract comprises its theory of face equivalence, its classification of unfaithful maps with at most two flag orbits, and its faithfulness and thinness theorems for mixes. Auxiliary pictured examples are not themselves a classification census, and their individual presentations are not regenerated here. Source errors are identified explicitly rather than attributed to new theorems of the common paper. The three final problems are recorded separately from this principal contract.

2 Color support and endpoint identifications

Lemma 1. *Let u and v be color words with disjoint supports. If $x \sim xuv$, then $x \sim xu$. If the two words commute, then also $x \sim xv$.*

Proof. For a color i outside the support of u, its path already shows xE_ixu. For a color in that support, v avoids i, so xuE_ixuvE_ix. This accounts for every color. When the words commute, interchange them and repeat the argument. □

Two i-adjacent flags have every other face in common. They are chain-equivalent precisely when they have their i-face in common as well. There is only one i-face precisely when this happens at every flag: if every i-edge stays in an i-face, no edge leaves that face, so connectedness makes it the whole graph. These facts give [F, Lemma 3.1 and Propositions 3.2–3.3], including the premaniplex case. A single face implies unfaithfulness whenever some edge of its color is not a semiedge; no such implication is valid for the one-flag premaniplex.

Lemma 2. *If $x \sim y$, then $xr_0 \sim yr_0$ and $xr_{n-1} \sim yr_{n-1}$. If the maniplex is unfaithful, then for each $1 \le i \le n-2$ there are distinct equivalent flags joined entirely by colors below i or entirely by colors above i.*

Proof. For the second assertion take distinct $x \sim y$. A path avoiding i can be written uv, where u uses colors below i and v uses colors above i. The two words commute. Lemma 1 puts both xu and xv in $C(x)$, and at least one differs from x, since $xuv = y \neq x$.

For the first assertion every face of rank other than zero already contains both xr_0 and yr_0. In particular a path between them avoiding color 1 has the form $r_0^\epsilon w$, where w uses only colors $2, \dots, n-1$ and $\epsilon \in \{0, 1\}$. If $\epsilon = 0$, this is also a path avoiding zero, which proves the claim. Otherwise $yr_0 = xw$. Since xE_0y, this gives yE_0yr_0. Multiplying the displayed identity on the right by r_0, and using commutation, also gives $y = xr_0w$, hence xE_0xr_0. Transitivity now gives $xr_0E_0yr_0$. Duality proves the other endpoint assertion. □

In rank three, the middle face consists of the four flags x, xr_0, xr_2, xr_0r_2. Lemma 1 therefore shows that a map is unfaithful exactly when some flag is equivalent to its 0-neighbor or its 2-neighbor. If $x \sim xr_1$, simplicity forces $xr_1 = xr_0r_2$; the same lemma then puts all four flags in $C(x)$. This proves [F, Propositions 3.4–3.9].

The rest of [F, Section 3] also follows directly from face relations. Parallel edges force their endpoints to be equivalent, whereas a path with possible semiedges is faithful: its unique path between distinct vertices cannot avoid every color. This gives the stated two- and three-flag premaniplex tests. Covers preserve equivalence by mapping color-avoiding paths to such paths. Equivalence inside a facet implies equivalence in the ambient graph, since its witnessing paths also avoid the new color. Thus ambient classes are unions of facet classes. If every class has size at most two, the same holds in each facet. If there is just one facet, and two distinct flags are equivalent *inside that facet*, they must be joined by the new-color edge: both the old pair and that edge lie in an ambient class of size at most two. This is the needed hypothesis in [F, Proposition 3.21]; merely being joined by a path in the facet does not imply facet equivalence. Finally two unique faces of different ranks give at least three distinct equivalent flags x, xr_i, xr_j, ruling out classes of size at most two.

3 Symmetry criteria without an orbit bound

Let $G \le \mathrm{Aut}(M)$. The common transport theorem applies to its free action on the flags. Here $H_i(x)$ is precisely the setwise stabilizer of $F_i(x)$. Its diagonal case gives [F, Theorem 4.2 and Corollary 4.3]. If $y \sim x$, it gives

$$yg \sim x \quad \Longleftrightarrow \quad g \in H(x), \qquad H(x) = \bigcap_i H_i(x).$$

This is [F, Theorem 4.6]; its disjoint union of torsors gives Corollaries 4.7–4.8, including the bound by the number of flag orbits when $H(x) = 1$. For different orbits, the full transporter formula and quotient-path lift in the common paper give exactly Theorem 4.15 and Corollary 4.16. No appeal to a finite enumeration is necessary.

Suppose now that there are two flag orbits, with I_0 the set of colors preserving each orbit. Let $i, j \notin I_0$ be distinct. For a base flag x, define τ_{ji} by $x\tau_{ji} = xr_jr_i$. An automorphism with this value exists and is unique, since the two color changes return to the base orbit. The transporters from xr_i to x are $H_k(x)$ for $k \neq i$ and $\tau_{ji}H_i(x)$ for $k = i$. Indeed, deleting a color other than i permits the i-edge; for color i, the path using j supplies the stated representative. Consequently

$$xr_ig \sim x \quad \Longleftrightarrow \quad g \in \tau_{ji}H_i(x) \cap \bigcap_{k \neq i} H_k(x). \tag{1}$$

This is [F, Proposition 4.11]. Automorphism equivariance makes class sizes constant on each orbit. Lemma 2 makes them equal across the two orbits whenever an endpoint color changes orbit, proving Proposition 4.9.

For an orientable rotary maniplex, apply (1) to its rotation group and $i = 0, j = 1$. If the distinguished rotations are σ_i, with $x\sigma_i = xr_ir_{i-1}$, the criterion is

$$H(x) = 1, \qquad S := \sigma_1 H_0(x) \cap \bigcap_{k=1}^{n-1} H_k(x) = \varnothing. \tag{2}$$

This includes orientable regular maniplexes by using their index-two rotation subgroup. If $\langle\sigma_i\rangle \cap \langle\sigma_j\rangle$ has a nonidentity element for $|i - j| \geq 2$, its two word representatives together avoid every color, placing it in $H(x)$ and disproving faithfulness. Finally $1 \in S$ says $x \sim xr_0$, and $\sigma_1 \cdots \sigma_{n-1} \in S$ says $x \sim xr_{n-1}$, because $xr_0\sigma_1 \cdots \sigma_{n-1} = xr_{n-1}$. Rotary symmetry and the corresponding endpoint adjacency propagate each equality to every flag. Thus these conditions say respectively that there is a single vertex or facet. This proves [F, Corollaries 4.12–4.13 and Proposition 4.14] in the orientable case. For a nonorientable regular maniplex, Proposition 4.14 alone follows directly by $H_i = \langle\rho_j : j \neq i\rangle$ and $\sigma_i = \rho_{i-1}\rho_i$. In that case the rotation group is the whole group, so the orientable criterion (2) is not asserted. In particular $1 \in \sigma_1 H_0$ is equivalent to $\rho_0 \in H_0$, and $\sigma_1 \cdots \sigma_{n-1} = \rho_0\rho_{n-1} \in H_{n-1}$ is equivalent to $\rho_{n-1} \in H_{n-1}$; the other memberships are automatic when the corresponding endpoint subgroup is the whole group.

4 One-face maps and dihedral centralizers

For a regular unfaithful map, an endpoint identification at one flag holds at every flag. Up to duality there is therefore one facet, so the flag set is the regular polygon action of $D = \langle a, b\rangle$, where $a^2 = b^2 = 1$. Let p be the order of ab; it is the number of edges of the facet, and may initially be infinite. The third mark c commutes with a and differs from $1, a, b$. The centralizer of a reflection in a finite dihedral group is $\{1, a\}$ when p is odd and $\{1, a, h, ha\}$ when p is even, where $h = (ab)^{p/2}$; in the infinite group it is $\{1, a\}$. These assertions follow by writing an element as $(ab)^m$ or $(ab)^m a$ and using $a(ab)a = (ab)^{-1}$. Thus $p = 2k$ is finite and $c = h$ or ha. These are precisely the regular maps called $P\{k\}$ and $T\{k\}$ in [F, Proposition 5.1], and their duals. Parameters producing repeated marks are omitted: in particular $T\{1\}$ is not a maniplex. The underlying group has order $4k$; taking the orders of bc gives the source's types. For $c = h$, this order is 2. For $c = ha$, it is k when k is odd and $2k$ when k is even.

Now suppose there are exactly two flag orbits. If a color endpoint changes orbit, its own edge or the other endpoint edge propagates an endpoint identification to both orbits, by Lemma 2. Hence an unfaithful two-orbit map has a single vertex or facet unless its symmetry type is $2_{\{0,2\}}$. This proves [F, Proposition 5.2].

The one-facet case can be treated without a separate presentation for every symmetry type. Its automorphism group is an index-two subgroup H of the polygon group D: the full polygon group acts regularly on flags, and H has two orbits. Write its character as $\chi : D \to C_2$. A color connection commutes with the left H-action and is determined by its values at one flag in each orbit. Let a, b again be the polygon's color 0 and color 1 connections.

If $\chi(a) = \chi(b) = 1$, then H is the rotation group. Let w be the third neighbor of the identity flag. Commutation with a determines the connection on the second orbit. If $w \in H$, involutivity forces $w^2 = 1$; if $w \notin H$, it forces $(wa)^2 = 1$. In either case the only allowed nontrivial rotation is the central half-turn. The third connection is then one global right multiplication, so the full polygon group acts and the map is regular. This excludes the two-orbit types $2_\varnothing$ and $2_{\{2\}}$ with one facet. If $\chi(a) = 0, \chi(b) = 1$ and the third color changes orbit, its value $w \notin H$ must commute with a. The same centralizer calculation gives $w = h$ or ha. Involutivity then makes the connection global right multiplication, again giving regularity. This excludes type $2_{\{0\}}$ with one facet. Duality handles the corresponding one-vertex cases. Together with the previous paragraph this proves [F, Propositions 5.3–5.4].

5 The other one-facet types

It remains to take $\chi(a) = 1, \chi(b) = 0$. The polygon must have $p = 2k$ edges, or be infinite; color its vertices alternately black and white. Its two H-orbits are those two flag colors. Choose an involution $w \in H$, and define the third connection by

$$R_2(x) = \begin{cases} xw, & x \text{ black}, \\ xawa, & x \text{ white}. \end{cases} \tag{3}$$

The first two connections are a, b. The rule commutes with a because that connection interchanges the colors. It is an involution because w preserves color. It is fixed-point free exactly when $w \neq 1$, never coincides with a, and coincides with b precisely when $w = b$ or aba. All these conclusions use the free polygon action and therefore hold for infinite polygons as well.

Every color-preserving polygon automorphism commutes with (3). A color-changing one can exist precisely when the automorphism mapping the base flag to its 0-neighbor exists: the preserving subgroup already acts transitively on each color. This condition is $aw = wa$. In the finite case its additional possibilities are $h = (ba)^k$ and ah; in the infinite case there are none. Thus (3) gives exactly the unfaithful maps of type $2_{\{1,2\}}$ when

$$w \notin \{1, b, aba, h, ah\}, \qquad w \in \ker \chi, \quad w^2 = 1. \tag{4}$$

The terms h, ah are omitted for an infinite polygon. Conversely any map of this type has one facet: it has two vertex orbits, so cannot have a single vertex. Equivariance determines its third connection on the black flags by one element w; commutation with a determines it on the white flags. This proves (3)–(4), and hence [F, Proposition 5.6].

Duality gives type $2_{\{0,1\}}$. Replacing color 2 by the product of colors 0 and 2 is the opposite operation. Applied to (3), it gives type $2_{\{1\}}$ with one facet. Its third connection now changes color, so cannot coincide with b; the exclusions b, aba in (4) are no longer needed. The exclusion of 1 prevents coincidence with a, and h, ah still exclude regularity. Duality gives the one-vertex case, equivalently the Petrie dual of the dual of (3). These operations are involutive, so their converses follow by reversing them. This proves all four parts of [F, Theorem 5.8], with the infinite-parameter convention made explicit. The literal occurrence of a in [F, Remark 5.7] as a color-preserving exceptional choice is inapplicable: $\chi(a) = 1$; the possible repeated-edge choice there is b.

6 The remaining symmetry type

The complete construction and proof for type $2_{\{0,2\}}$ are given in the companion *A corrected construction for unfaithful maps with two flag orbits*. Its input is a regular map with finite facets of even length. In coordinates $G \times \{0, 1\}$, let its marks be a, b, c, let $p = |ab|$ be even, and put $h = (ab)^{p/2}$. The new marks are

$$R_0(g, 0) = (ga, 0), \quad R_0(g, 1) = (gb, 1), \quad R_1(g, e) = (g, 1 - e),$$
$$R_2(g, 0) = (gc, 0), \qquad R_2(g, 1) = (gd, 1), \quad d = h \text{ or } hb. \tag{5}$$

These constructions and their duals exhaust the unfaithful maps of the remaining type, with the regular cases removed. The source's printed torus rule uses ha in place of hb. For $p > 2$, ha does not commute with the white color-0 mark b; for $p = 2$ it equals b, giving parallel edges. Thus that literal rule never gives a maniplex on a valid polygon facet. The companion proves the corrected criterion and explains the swap of the two flag orbits needed when an initial base marking repeats a generator. This is a correction to the source's formulation, not a claimed solution of one of its open problems. The following unnumbered source paragraph about the exact sizes of all chain-equivalence classes is outside this recovery: the construction's validity and exhaustive converse do not by themselves prove that assertion.

7 Covers, mixes and thinness

A cover of a faithful maniplex that is injective on each facet is faithful by the second part of the common paper's joint-map theorem. This is [F, Proposition 6.1]. The coordinate projections of a mix jointly distinguish its flags and preserve face equivalence. Thus

$$C_{X\diamond Y}(x,y) \subseteq C_X(x) \times C_Y(y).$$

The coordinate pairs in the first sentence of [F, Proposition 6.2] are transposed: the intended implications are $x \sim x'$ and $y \sim y'$, for equivalent pairs $(x,y),(x',y')$. The displayed inclusion and subsequent corollary in the source have the correct meaning. The common theorem now gives its Corollaries 6.3–6.4, and even removes the finite-orbit restriction from the latter.

The common paper's deleted-face criterion is precisely the equivalence in [F, Propositions 7.1–7.2]. Applying it to the mix, the two-coloring forces both coordinates to choose the same flag or the same adjacent flag. This gives [F, Theorem 7.3 and Corollary 7.4]. No claim is made that orientability can be dropped. The source's cited nonorientable example remains an attributed example, not an additional finite census carried out here.

8 What is recovered and what remains separate

The generic transporter theorem recovers all the symmetry criteria without a bound on the number of orbits. The support lemma and endpoint propagation reduce the rank-three classification to the centralizer of a reflection. The common paper also proves all covering and mixing implications used here. These are complete written arguments; they do not rely on sampled finite groups. The companion supplies the last construction with its corrected literal formula and full converse.

The attribution remains with the authors of [F] for the original results, and with the earlier authors cited there for the rank-three endpoint criterion and classical one-face maps. The new organizational step is to use one common refinement and transporter calculation throughout. Standard facts used in that calculation are proved in the common paper. The additional arbitrary-family result is not assigned historical novelty. The reconstruction criterion is kept separate from recovery of prior results: it treats natural reconstruction in Problem 3, and arbitrary isomorphism for finite posets, with the remaining interpretation stated explicitly. Problems 1 and 2 are not claimed solved. Nor does the present application claim complete Lean formalization of the classification.